

Reflection Journal

When I first started looking at the end-to-end Machine Learning (ML) workflow, it honestly felt like a lot to take in. You're jumping between loading CSV files, calculating stats, filling in missing data, building Seaborn charts, and setting up training pipelines. At first, it just seemed like an intimidating checklist of steps to get through. But once I started working through the Titanic dataset, things really clicked: I stopped seeing ML as a rigid list of tasks and started seeing it as a logical, step-by-step story where every phase directly shapes the next.

I realized that Exploratory Data Analysis (EDA) isn't just something you do to make pretty charts before the real work begins. It's the foundation of the whole process. You can't build a reliable predictive model without first understanding the layout, weird quirks, and hidden connections in your data. Running initial checks with `df.info()` and `df.describe()` brought some huge details to light. For instance, seeing that 177 values were missing in Age and that Cabin was mostly empty (only 204 non-null entries out of 891) completely changed how I thought about feature engineering. Instead of guessing how to fix missing data, I could plan a targeted strategy. Without taking the time to diagnose these issues upfront, feeding messy data straight into a model would just lead to bad predictions, classic "garbage in, garbage out."

My biggest lightbulb moment came when I compared basic summary statistics to visual charts. At first, I thought simple numbers like the mean, standard deviation, and quartiles were enough to understand a dataset. But analysing the Titanic passenger data proved just how much single summary numbers can hide.

Take passenger age, for example. The overall average age came out to 29.7 years. If you just looked at that number, you'd probably picture a standard bell curve centered around young adults. But when I built side-by-side histograms using FacetGrid to compare survivors and non-survivors, a huge detail popped out. There was a massive spike in survival for infants and young kids under 10. That critical detail completely disappears if you only look at the average age.

We saw the exact same thing with passenger fares. The mean fare was \$32.20, but the raw numbers showed a minimum of \$0.00 and an extreme maximum of \$512.33. Creating a boxplot of Fare versus Survived made the heavy skew obvious right away. The median fare for people who died was around \$10.00, while the distribution for survivors stretched way higher, with an interquartile range going past \$50.00 and plenty of high-value outliers. A single average fare of \$32.20 gave a totally misleading picture for both groups. Visualizing the data made everything clear and showed how much socioeconomic status influenced who got access to lifeboats.

Digging into categorical features gave me a much better grasp of how feature interactions drive classification models. Plotting the target variable with `sns.countplot()` laid out the baseline reality: 549 passengers died while 342 survived, giving a baseline survival rate of 38.4%. Spotting this class imbalance early is key for model design. If you built a simple model that just guessed "Died" for every single person, it would claim an accuracy of 61.6%, even though it completely failed at predicting anything useful.

Breaking survival down by different variables brought out some really strong patterns:

1. Gender (Sex): Over 70% of female passengers survived, compared to under 20% of male passengers, making gender the strongest individual predictor in the dataset.
2. Passenger Class (Pclass): More than 60% of 1st Class passengers survived, while 3rd Class passengers saw a survival rate under 25%. This clearly showed how ticket class played a role during the evacuation.
3. Port of Embarkation (Embarked): Passengers who boarded at Cherbourg (C) had a noticeably higher survival rate than those who got on at Southampton (S) or Queenstown (Q). But looking closer, this wasn't really about geography, it turned out Cherbourg simply had a much higher percentage of 1st Class passengers.